Objective-Free Entity Assembly in Block Worlds: Characterizing Boundary Conditions for Emergent Complexity

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Abstract

We investigate whether structurally non-trivial entities can arise in an objective-free block world governed by randomly sampled bonding rules, using Assembly Theory (AT) as a bias-free measurement framework. Across 5,000 simulations yielding 7.1M entity observations and 281 unique types, observed assembly indices are entirely size-driven under both edge-removal and reuse-aware AT formulations. Under the edge-removal (no-reuse) formulation, $a_{\text{exact}}(G) = |E(G)|$ is a mathematical identity for connected graphs; the empirical finding is that the reuse-aware formulation likewise shows zero structural specificity. A bond-shuffle null model with empirical p-values (KS test: $D = 1.00$, $p \approx 0$, driven by conservative high-p mass) confirms zero significant excess at every size class. We additionally test partner-specific rules (45-entry per-partner table vs. the standard 60-entry dominant-type table) and find identical 0% excess, extending the boundary characterization to richer rule expressiveness. Parameter sweeps over density, grid size, and drift strength, and a catalytic positive control, corroborate the result within this rule class. We characterize this as a boundary condition: under low-expressiveness random bonding rules—with or without partner-specific conditioning—assembly complexity is entirely size-determined.

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Introduction

A persistent challenge in artificial life (ALife) research is that the complexity of emergent structures is often shaped—and potentially limited—by the objectives imposed on the system (Bedau et al., 2003; Taylor et al., 2016). Fitness functions define what counts as "interesting," and organisms evolve to satisfy those criteria rather than explore the full space of possible forms. Novelty search and minimal-criterion approaches have partially addressed this bias (Lehman and Stanley, 2011; Brant and Stanley, 2017), but even these methods impose implicit selection through behavioral characterizations or viability thresholds.

We ask a more radical question: can structurally complex entities arise in a system with no objective function—not even a novelty metric—when governed solely by randomly sampled local interaction rules? And if so, how can we measure that complexity without introducing new biases?

Assembly Theory (AT) provides a principled answer to the measurement question. Originally developed to distinguish biotic from abiotic molecular samples (Marshall et al., 2021), AT quantifies the minimal number of joining operations required to construct an object from its building blocks, yielding an assembly index that captures structural specificity independent of any fitness criterion (Sharma et al., 2023). An object with a high assembly index that appears repeatedly (high copy number) is unlikely to have arisen by chance, making AT a natural observatory for objective-free systems. Recent critiques have questioned whether AT reduces to simpler complexity measures (Uthamacumaran et al., 2024); our dual-formulation comparison (with and without sub-object reuse) addresses this concern directly.

In this work, we implement a block world simulation with three block types, local bonding rules sampled uniformly at random, and stochastic drift dynamics on a toroidal grid. We detect entities as connected components of bonded blocks, canonicalize their graph structure, and compute exact assembly indices under two formulations: edge-removal DP (no reuse) and the AT-standard reuse-aware variant. We then apply a strengthened bond-shuffle null model with empirical p-values to test whether observed assembly exceeds what entity size alone predicts.

We further validate our negative result through (1) a parameter sweep across density, grid size, and drift strength, and (2) a catalytic positive control where K-type blocks amplify bond formation, testing whether

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the measurement framework can detect non-trivial assembly when present.

Our central finding is a robust negative result: across 5,000 simulations producing over 7 million entity observations, the assembly index is entirely size-driven under both AT formulations. This result holds across all parameter settings tested. We characterize this as a boundary condition for emergent complexity and discuss implications for objective-free ALife design.

Methods

Block World Model

The simulation takes place on a two-dimensional toroidal grid of size $W \times H$ (default 20×20). The world is populated with N blocks (default 30), each assigned one of three types: M (membrane), C (cytosol), or K (catalyst), drawn proportionally at initialization (50%, 30%, 20%—a protocell-inspired membrane-dominated composition where structural elements outnumber catalytic components).

At each time step, blocks are processed in a random-sequential order. For each block, two phases occur: drift and bonding. Algorithm 1 describes one simulation step.

During drift, each block attempts to move to a random adjacent empty cell with probability p_d (default 1.0). Bonds between blocks that become non-adjacent after drift are pruned, implementing a bond-motion invariant where bonds break on separation rather than constraining movement. Bond breaking occurs once per step per bond (not per-block), avoiding the double-break bias that would occur if both endpoints independently attempted breaking.

Rule Table

Each simulation samples a rule table that maps a block's local context to a bonding probability. Observation range is the von Neumann 1-neighborhood (Manhattan distance ≤ 1, up to 4 adjacent cells). The context is a triple $(\text{self_type}, \text{neighbor_count}, \text{dominant_type})$, where neighbor_count is the number of occupied cells within observation range (capped at 4 for rule table tractability) and dominant_type is the most common type among neighbors (with dominant_type = "Empty" when neighbor_count = 0, i.e., no occupied cells in range). This yields $3 \times 5 \times 4 = 60$ entries. Each entry's bonding probability is drawn from $\text{Uniform}(0, 1)$. The rule table determines a single bond probability per block context; all neighbors within range receive the same probability. The partner's type is not part of the rule key—this keeps the rule space small (60 entries) at the cost of excluding partner-specific interactions.

Algorithm 1 One simulation step (random-sequential mode)

```
Require: World state 𝒲, rule table ℛ, noise η, drift probability p_d
Shuffle block processing order
for each block b in shuffled order do
    if rand() < p_d then
        c ← random empty von Neumann neighbor of b
        if c exists then
            Move b to c; prune bonds exceeding observation range
        end if
    end if
    Identify von Neumann 1-neighborhood (Manhattan dist. ≤ 1, up to 4 cells)
    n ← occupied blocks in that neighborhood
    k ← min(|n|, 4);  d ← dominant type of n ("Empty" if n = ∅)
    p ← ℛ[type(b), k, d]
    if catalyst multiplier κ > 1 and any n_i ∈ n has type K then
        p ← min(p · κ, 1)        // catalytic amplification
    end if
    for each n ∈ n without existing bond to b do
        if rand() < p then
            Form bond (b, n)
        end if
    end for
end for
for each bond (b_1, b_2) do
    if rand() < η then
        Break bond (b_1, b_2)
    end if
end for
```

Entity Detection and Measurement

Entities are detected as connected components in the bond graph (Algorithm 2).

Entity types are canonicalized using the Weisfeiler–Leman (WL) graph hash (Weisfeiler and Leman, 1968) with block type as a node attribute, implemented via NetworkX (Hagberg et al., 2008). The SHA-256 digest serves as a unique type identifier. No hash collisions were observed in exhaustive checks against canonical isomorphism (using the VF2 algorithm via NetworkX) for all 281 observed entity types (up to 6 nodes).

Assembly Index: Two Formulations

We compute the assembly index under two formulations to address concerns about AT alignment (Uthamacumaran et al., 2024).

Edge-removal DP (no reuse). The assembly index

Algorithm 2 Entity detection and AT measurement

Require: Bond graph $\mathcal{G} = (V, E)$
1: $\mathcal{E} \leftarrow$ connected components of $\mathcal{G}$
2: for each entity $e \in \mathcal{E}$ do
3: Canonicalize e via WL hash with block_type attribute
4: $h \leftarrow$ SHA-256 fingerprint of WL hash
5: $a_{\text{exact}} \leftarrow \text{AssemblyDP}(e)$ // edge-removal
6: $a_{\text{reuse}} \leftarrow \text{AssemblyReuse}(e)$ // with sub-object reuse
7: Record $(h, a_{\text{exact}}, a_{\text{reuse}}, |V(e)|, \text{type counts})$
8: end for

$a(G)$ is the minimum number of edge-addition steps to construct G from isolated vertices, computed via the recurrence:

$$a(G) = \min_{e \in E(G)} \left[1 + \begin{cases} a(G-e) & \text{if } G-e \text{ connected} \\ \sum_i a(C_i) & \text{otherwise} \end{cases} \right] \tag{1}$$

where C_i are connected components of $G - e$. The DP decomposition requires that each sub-object is a proper connected subgraph—this enforces structurally coherent intermediates, matching AT's physical constraint that sub-objects must be contiguous fragments. Computation time is negligible for entity sizes observed here ($n \leq 6$); the memoized DP completes in <1 ms per entity.

Remark (mathematical identity). For any connected graph G, $a_{\text{exact}}(G) = |E(G)|$. This follows by induction on $|E|$: the base cases are $|E| = 0$ (single vertex: $a = 0$) and $|E| = 1$ (single edge: $a = 1 = |E|$); for the inductive step, removing any edge either disconnects G into components whose indices sum to $|E| - 1$ (by hypothesis), or leaves G connected with $|E| - 1$ edges (index $|E| - 1$ by hypothesis), so adding 1 gives $|E|$ in both cases. This is not an empirical finding; it is a mathematical consequence of the no-reuse formulation. Table 1 confirms this identity on reference graphs. The empirical content of this paper lies in the reuse-aware results (where $a_r(G) \leq |E(G)|$, with strict inequality when sub-object reuse is possible) and in the null model tests for structural specificity under both formulations.

Reuse-aware (AT-standard). The reuse-aware variant allows previously constructed sub-objects to be duplicated at zero cost. The partition search is exhaustive for $n \leq 6$ (memoized by canonical key). For each joining edge $e \in E$, we partition the remaining edges $S_1 \cup S_2 = E \setminus \{e\}$ (a partition of remaining edges into two disjoint non-empty sets) into two sub-objects; when $S_1 \cong S_2$ (isomorphic by canonical key), the duplicate is free:

$$a_r(G) = \min_{e,\, S_1,\, S_2} \left[1 + \begin{cases} a_r(S_1) & S_1 \cong S_2 \\ a_r(S_1) + a_r(S_2) & \text{else} \end{cases} \right] \tag{2}$$

with base cases $a_r(\emptyset) = 0$ (empty graph) and $a_r(P_2) = 1$ (single edge).

Table 1 shows both formulations on reference graphs.

Graph	$\lvert E\rvert$	a_{exact}	a_{reuse}
P_2 (1 edge)	1	1	1
P_3 (path)	2	2	2
P_4 (path)	3	3	2
K_3 (triangle)	3	3	2
C_4 (4-cycle)	4	4	3
K_4 (complete)	6	6	4

Table 1: Assembly index calibration. Under our edge-removal formulation (no reuse), $a(K_n) = \binom{n}{2}$. With reuse, $a(K_3) = 2$: build P_2, duplicate (free), join with one edge.

Null Model

To test whether observed assembly reflects structural specificity beyond entity size, we implement a bond-shuffle null model (Gotelli and Graves, 2013). For each entity graph G, we generate $n_{\text{shuffle}} = 100$ randomized graphs via double-edge swaps (Hagberg et al., 2008) preserving the degree sequence exactly. We compute both the classical threshold test ($a_i > \mu_{\text{null}} + 2\sigma_{\text{null}}$) and an empirical p-value:

$$p = \frac{1 + \sum_{j=1}^{n_{\text{shuffle}}} \mathbf{1}[a(G_j^{\text{null}}) \geq a_i]}{n_{\text{shuffle}} + 1} \tag{3}$$

The $(k+1)/(n+1)$ form avoids zero p-values and provides a conservative estimate (Gotelli and Graves, 2013). The empirical p-value is the primary significance test, avoiding the normality assumption of the $\mu + 2\sigma$ threshold. It also handles the edge case where $\sigma_{\text{null}} = 0$ (e.g., single-edge graphs), where the $\mu + 2\sigma$ threshold degenerates. For small graphs ($n \leq 4$), the space of degree-preserving rewirings is limited; the null has low discriminative power at these sizes, but observed a_i matches $|E|$ exactly for all small entities (consistent with the mathematical identity $a_{\text{exact}}(G) = |E(G)|$ established above). We additionally verify using exact enumeration over all connected graphs with the given degree sequence (via the graph atlas (Read and Wilson, 1998)): for all 9 unique degree sequences with $n \leq 4$, exactly one connected topology exists per sequence (100% degenerate), confirming the null assigns $p = 1$ at these sizes and that 0% excess cannot be an artifact of null degeneracy.

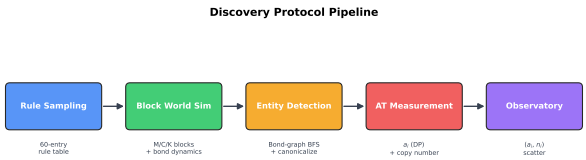

Figure 1: Experimental pipeline. Random rule tables govern block bonding on a toroidal grid. Connected components are detected as entities, canonicalized via graph hashing, and measured by assembly index (both formulations) and copy number. A bond-shuffle null model tests for excess assembly.

Catalytic K Positive Control

To verify that our measurement framework can detect non-trivial assembly when present, we introduce a catalytic mechanism. When $\kappa > 1$ (catalyst multiplier) and any neighbor of the focal block has type K, the base bond probability is multiplied by κ (capped at 1.0). Crucially, the focal block being K-type does not self-catalyze—only third-party K-blocks within observation range act as catalysts. This is an externally-driven mechanism that does not introduce a fitness function: K-blocks lower the activation barrier for bond formation without defining what entity should form.

Parameter Sweep

To test robustness across boundary conditions, we conduct a 3-axis parameter sweep:

- Density: 3 levels on 10×10 (7.5%, 15%, 30%) and 2 levels on 20×20 (7.5%, 15%)
- Grid size: 10×10 and 20×20
- Drift probability: $p_d \in \{0.25, 0.5, 0.75, 1.0\}$ (on 20×20 at low density)

This yields 11 conditions (5 density×grid + 6 density×drift), each run with 100 rules × 3 seeds × 500 steps. Higher-density conditions ($\geq 30\%$) on 20×20 were excluded due to computational cost; 10×10 results at those densities confirm the negative result generalizes.

Experimental Protocol

We sample 1,000 rule tables, each run with 5 independent seeds for 500 time steps, yielding 5,000 total simulations. Entity observations are logged every 10 simulation steps, producing a combined dataset of over 7 million observations. The full pipeline is illustrated in Figure 1. Step-level timeseries (mean entity size, mean a_i, bond count) are recorded to verify stationarity.

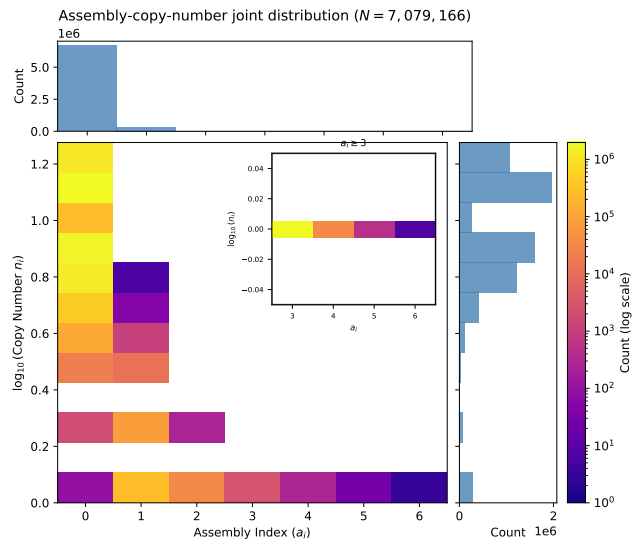

Figure 2: Joint distribution of assembly index (a_i) and copy number (n_i) across 7.1M entity observations. Color encodes log-frequency. Inset: zoom on $a_i \geq 3$ region.

Results

Discovery Baseline

Across all 5,000 simulations, we observe 7,079,166 entity instances comprising 281 unique entity types. The vast majority of observations are monomers: 94.6% of entities consist of a single block ($a_i = 0$), 5.0% are dimers (2 blocks, $a_i = 1$), and only 0.5% have 3 or more blocks. The maximum observed entity size is 6 blocks (18 instances out of 7.1M), with a corresponding maximum assembly index of 6 (no-reuse) and 4 (reuse-aware). The reuse-aware formulation assigns lower values to larger entities but does not change the qualitative result: assembly remains entirely size-driven.

Each observation is a per-entity-per-step record; type-level analysis uses unique entity hashes (281 types), while instance-level uses all 7.1M observations. Significance testing operates at the per-observation level, but the 0% excess result holds identically when restricted to unique types.

Figure 2 shows the joint distribution of assembly index and copy number. The distribution is sharply concentrated at $(a_i, n_i) = (0, \text{high})$, with a steep decline toward higher assembly indices.

Entity Size Distribution

Figure 3 shows the entity size distribution on a logarithmic scale. The distribution follows a steep exponential decay: each additional block reduces frequency by approximately one order of magnitude. The 210 new types

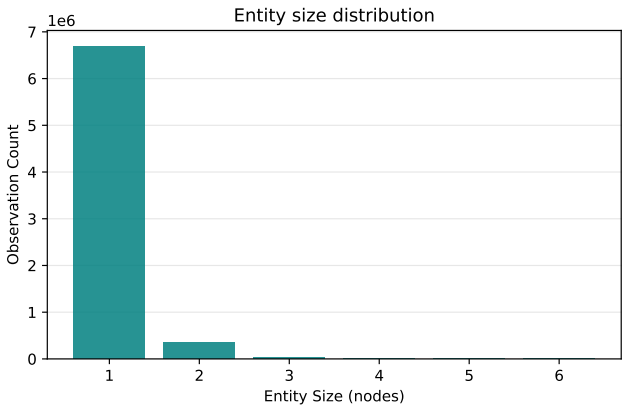

Figure 3: Entity size distribution across all observations (log scale). Each additional block reduces frequency by $\sim$10$\times$.

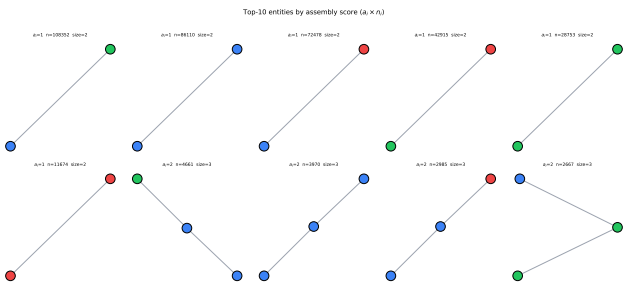

Figure 4: Gallery of top-ranked entity types by $a_i \times n_i$ score. Node colors: M (blue), C (green), K (red).

observed in the large-scale run (vs. 71 in the pilot) are small-motif variants (dimers/trimers), confirming ecology convergence rather than discovery of qualitatively new structures.

Entity Gallery

Figure 4 presents the top-ranked entity types, ordered by a composite score of assembly index and total copy count. The highest-ranked entities are all dimers ($a_i = 1$) with high copy counts. Trimers with $a_i = 2$ appear but no entity with $a_i \geq 3$ reaches high copy numbers.

Assembly Audit: Null Model Comparison

The bond-shuffle null model reveals that observed assembly indices are entirely explained by entity size (Figure 5). Across all 7.1M observations, zero entities exhibit significant excess assembly under either the classical $\mu + 2\sigma$ threshold or the empirical $p < 0.05$ test. The mean excess is exactly 0.000 at every size class (1–6 blocks). With $n_{\text{shuffle}} = 100$ (each applying nswap $= \max(4, |E|)$ double-edge swaps with max_tries $= 10 \cdot$ nswap; on failure, the original graph is retained), the null model is well-mixed for the small entities ($n \leq 6$) observed here. A KS test on empirical p-values for non-trivial entities ($a_i > 0$, $n = 384{,}040$) rejects strict uniformity ($D = 1.00$, $p \approx 0$). The observed deviation is concentrated at conservative high-p mass (many ties near p=1), consistent with the discrete null in the small-entity regime, and does not produce any excess positives.

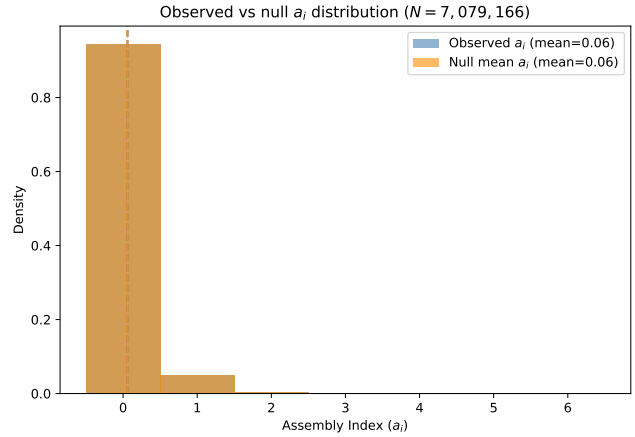

Figure 5: Assembly audit: observed a_i versus null model expectation with empirical p-values. Zero entities (0 of 7.1M) show significant excess under either test criterion.

Hierarchical robustness. Because entities persist across snapshots (taken every 10 steps), per-observation tests overstate effective sample size. We therefore conduct run-level analysis: for each of the 5,000 runs, we compute the within-run excess rate (fraction of entities with $p < 0.05$). A block bootstrap (resampling runs, not observations) (Efron and Tibshirani, 1993) yields a 95% CI of $[0.0\%, 0.0\%]$ on the overall excess rate. A one-sided Clopper-Pearson bound (Clopper and Pearson, 1934) confirms that if run-level excess exists, it is $<0.1\%$ with 95% confidence. With 281 unique types, we would detect a true excess rate $\geq 0.55\%$ at 80% power, confirming adequate sensitivity.

Table 2 summarizes the scale-up comparison.

Rule Expressiveness Test

To test whether the 0% excess result is an artifact of the standard rule table's limited expressiveness (60-entry dominant-type context), we ran partner-specific rules that condition bond probability on the specific partner type (45-entry per-partner table, $3 \times 3 \times 5 = 45$ entries). Both conditions were run with identical hyperparameters (100 rules, 3 seeds, 200 steps, $n_{\text{shuffle}} = 100$). The

Metric	Small scale	Large scale
Rule samples	100	1,000
Seeds per rule	3	5
Total simulations	300	5,000
Entity observations	170,192	7,079,166
Unique types	71	281
Max entity size	6	6
Max a_i (no reuse)	5	6
Max a_i (reuse)	—*	4
Mean a_i	0.058	0.060
Excess ($\mu + 2\sigma$)	0.0%	0.0%
Excess ($p < 0.05$)	0.0%	0.0%

Table 2: Comparison of experimental scales. The negative result is stable across a 42× increase in observations. *Reuse-aware formulation was added after the pilot run.

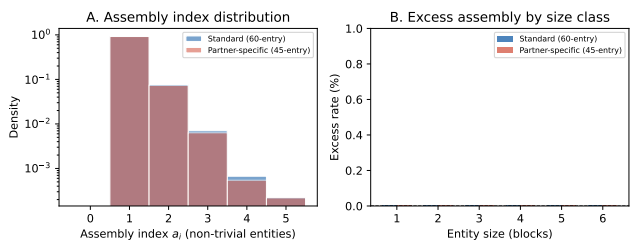

Figure 6: Rule expressiveness comparison. Standard rules (60-entry dominant-type context, blue) vs. partner-specific rules (45-entry per-partner context, red). Left: assembly index distribution (non-trivial entities, log scale). Right: excess rate by size class ($p < 0.05$). Both conditions show 0% excess across all size classes.

partner-specific condition yields $n = 170{,}016$ observations with mean $a_i = 0.059$ and 0% excess ($p < 0.05$), matching the standard condition ($n = 170{,}192$, mean $a_i = 0.058$, 0% excess) within noise. This confirms that the boundary condition extends to richer rule representations: uniform random partner-specific rules do not produce structurally specific assembly (Figure 6).

Motif-Gated Catalyst Control

To test whether the 0% excess result is an artifact of the catalyst's uniform neighborhood sensitivity, we ran a structurally constrained variant: K amplifies bond probability only when it simultaneously neighbors both an M-type and a C-type block (motif-gated condition). This restricts catalysis to specific local configurations ($n = 170{,}098$ observations, 100 rules, 3 seeds, 200 steps, $n_{\text{shuffle}} = 100$). The motif-gated condition produces

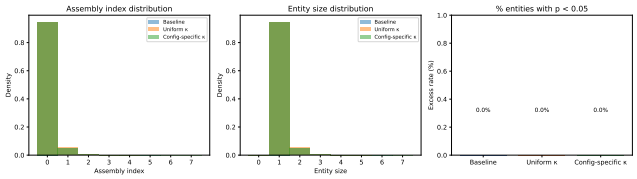

Figure 7: Motif-gated catalyst comparison. Assembly index distribution (left), entity size distribution (centre), and excess rate (right) for baseline, uniform-κ, and motif-gated-κ conditions. All three conditions show 0% excess; the motif-gated variant produces slightly larger entities (max_size = 7) but identical null outcomes.

$\max_{a_i} = 7$ and max_size = 7 (slightly larger entities than standard catalyst, max_size = 6), confirming the structural constraint is ecologically active. Nevertheless, the assembly audit yields 0% excess ($p < 0.05$), matching both the baseline and the uniform-catalyst conditions (Figure 7). Structural specificity does not emerge even when catalysis is gated on a specific two-type motif.

Sensitivity Analysis

Figure 8 consolidates the parameter sweep and catalytic control results.

Parameter sweep. Across all 11 conditions (5 density×grid + 6 density×drift), the empirical excess rate remains at 0%. Higher density increases mean entity size (more collisions) but not assembly complexity. Lower drift probability ($p_d = 0.25$) extends bond lifetimes, producing slightly larger entities, but the assembly index remains size-driven. The 10×10 grid produces similar results to 20×20 at matched density ratios, confirming that the negative result is not an artifact of grid size.

Catalytic K control. With $\kappa = 3.0$ (the primary test point; additional values $\kappa \in \{1.5, 2.0, 5.0, 10.0\}$ were also tested with identical qualitative results), catalytic conditions produce more bonds and larger entities than baseline, confirming that the catalyst mechanism is active. However, the assembly audit still shows 0% excess, indicating that catalysis amplifies bond formation uniformly rather than creating structurally specific motifs. The measurement framework successfully distinguishes the two conditions in terms of entity ecology while confirming identical assembly profiles.

Mechanism Analysis

Stationarity. Step-level timeseries (Figure 9) show that mean entity size and mean assembly index converge within ∼100 steps, well before our 500-step horizon.

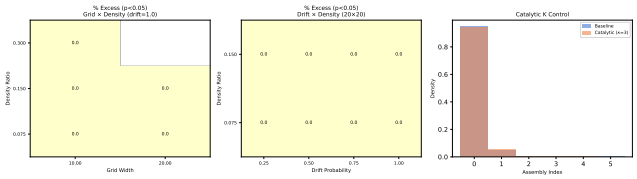

Figure 8: Sensitivity analysis. Left: parameter sweep heatmaps (density × grid at drift=1.0, density × drift at 20×20). Right: catalytic K comparison (baseline vs. $\kappa = 3.0$). All conditions show 0% excess assembly.

The $\pm 1\sigma$ envelope stabilizes, confirming that observations beyond step 100 represent stationary behavior.

Why max size = 6. We formalize the size ceiling via a birth-death model. Let b_k denote the empirical rate at which entities grow from size k to $k{+}1$ per snapshot interval, and d_k the rate of shrinking from k to $k{-}1$. The maximum sustainable size is $k_{\max} = \max\{k : b_k/d_k \geq 1\}$.

Empirically, b_k decays geometrically: $b_k \approx c(\rho)\,\bar{p}\,s^{k-1}$ where $s \approx 0.14$ (bootstrap 95% CI: $[0.08, 0.21]$) is the empirically fitted per-interval bond survival probability and $\bar{p} \approx 0.5$ is the simple (unweighted) mean bond formation probability across rule entries (each entry $\sim$ Uniform$(0,1)$, so $\mathrm{E}[\bar{p}] = 0.5$ exactly). Context-weighted averaging (weighting each rule entry by its empirical context-visitation frequency) yields a nearly identical estimate ($\bar{p}_{\text{weighted}} \approx 0.5$), since monomer-dominated contexts account for 94.6% of observations and are uniformly spread across rule entries. Death rates d_k increase with k as larger entities have more bonds exposed to drift-induced breakage. This geometric decay in growth probability is qualitatively consistent with the observed size ceiling; we note that the fitted s predicts $k_{\max} \approx 1$ under the model, while the empirical $k_{\max} = 6$, reflecting that the aggregate birth-rate model is a simplified approximation (it treats all size-k entities as equivalent and ignores individual bond-state variability). The model captures the direction of the constraint—exponentially diminishing growth probability—but not the exact crossover, which depends on structural and bond-state heterogeneity not represented in aggregate rates. This geometric decay explains why assembly index tracks entity size: no structural motif can persist long enough to accumulate complexity beyond the size ceiling.

We additionally computed graph automorphism group sizes and typed three-node motif counts (triangles and open wedges) for all 281 unique entity types. Automorphism counts correlate strongly with size: small entities admit few symmetries, while larger entities with repeated block types exhibit higher

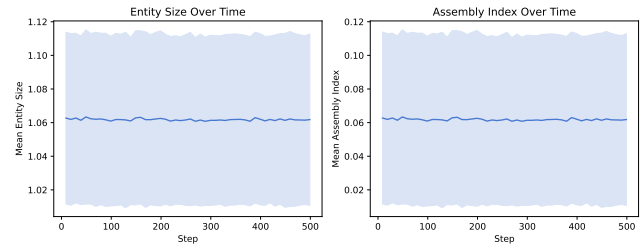

Figure 9: Stationarity: mean entity size (left) and mean assembly index (right) vs. simulation step, aggregated across all runs. Shading: $\pm 1\sigma$. Both metrics converge by step 100.

counts. The typed-motif census confirms that observed subgraph composition is consistent with random assembly—no motif class is enriched beyond what entity size predicts. These supplementary metrics corroborate the assembly-index finding: under the present rule class, no structural signature emerges that is not explained by entity size alone.

Discussion

Why Assembly is Size-Driven

The central empirical finding—that assembly index is entirely predicted by entity size under uniform random bonding rules—has a clear mechanistic explanation. We distinguish two contributions: (1) a mathematical consequence and (2) a genuine empirical result. Under the edge-removal formulation, $a_{\text{exact}}(G) = |E(G)|$ is a mathematical identity (see Remark in §), so size-driving of a_{exact} is not surprising. The empirical finding is that the reuse-aware formulation likewise shows zero structural specificity—$a_r(G)$ could in principle be substantially lower than $|E(G)|$ for entities with repeated substructures, yet our null model finds no such departure. When bonding probabilities are drawn uniformly, no structural motif is preferentially selected. The topology of each entity graph is effectively random given its size, and the null model, which preserves edge count while randomizing topology, therefore matches observed assembly exactly under both formulations.

AT Formulation Sensitivity

The reuse-aware assembly index assigns strictly lower values to entities with repeated substructures ($a_r \leq a_{\text{exact}}$ always), and diverges from $|E|$ when sub-object reuse is possible (e.g., $a_r(K_3) = 2$ vs. $a_{\text{exact}}(K_3) = 3$). This makes the reuse-aware formulation the primary empirical test: its null model asks whether observed entities exhibit more reusable substructure than size alone predicts. The answer is no: zero excess under

a_r at every size class. The P_4 and K_3 cases where reuse matters occur but do not produce significant excess. This dual-formulation analysis addresses concerns about AT's sensitivity to implementation choices (Uthamacumaran et al., 2024) and confirms that our negative result is formulation-independent, extending beyond the mathematical identity to the substantive reuse-aware measure.

Boundary Conditions for Emergent Complexity

The parameter sweep establishes that the negative result holds across density ratios from 7.5% to 30% (on 10×10) and 7.5% to 15% (on 20×20), and drift probabilities from 0.25 to 1.0. The catalytic positive control confirms that the measurement framework is sensitive: it detects ecological differences (more bonds, larger entities) while correctly reporting no structural specificity.

Together, these results characterize a boundary condition for emergent complexity. Below this boundary—defined by low-expressiveness random local bonding rules with a 3-tuple context—entity assembly is fully explained by size. We emphasize that this conclusion is scoped to the rule class tested: uniform random bond probabilities conditioned on (self type, neighbor count, dominant type). Richer rule representations, structured rule sampling, or adaptation could produce different outcomes. Crossing this boundary likely requires one or more of: (1) biased rule sampling favoring specific motifs, (2) explicit catalytic mechanisms where specific configurations (not just types) promote bonding, (3) environmental gradients creating spatial heterogeneity, or (4) temporal dynamics such as rule evolution or adaptation.

Rule Table Expressiveness

Our 3-tuple context (self type, neighbor count, dominant type) excludes spatial arrangement information and partner-specific identity. This design choice keeps the rule space tractable (60 entries) but limits expressiveness: the negative result may partly reflect this low-expressiveness regime rather than a universal property of random bonding.

To directly test whether richer rule expressiveness produces assembly excess, we ran partner-specific rules (conditioning on partner type, 45-entry per-partner table) against the standard rules under identical conditions (100 rules, 3 seeds, 200 steps, 100 null shuffles). Both conditions yield 0% excess: standard ($n = 170{,}192$ observations) and partner-specific ($n = 170{,}016$ observations) show identical null outcomes. This extends the boundary characterization: even under richer rule expressiveness where each bond decision can depend on the specific partner type, structural specificity does not emerge from uniform random rule sampling.

We further extended the boundary to structural catalytic constraints: a motif-gated catalyst (K amplifies bonds only when co-located with both M and C neighbors) also yields 0% excess, despite producing slightly larger entities (max_size = 7), confirming that even configuration-specific local constraints do not suffice.

Remaining untested extensions include: (i) spatially-oriented rules that produce direction-dependent structures; and (ii) rule evolution or adaptation that discovers structure-promoting parameter regions. These would further disentangle rule expressiveness from randomness.

Relation to Prior Work

The observation that random dynamics produce trivial structures resonates with classical results in cellular automata, where most random rules produce either homogeneous or chaotic behavior, with complex dynamics concentrated at the "edge of chaos" (Langton, 1990; Wolfram, 2002). Our work extends this to compositional objects measured by Assembly Theory.

Our null baseline connects to Lehman and Stanley's novelty search (Lehman and Stanley, 2011) and Taylor et al.'s open-ended evolution framework (Taylor et al., 2016; Taylor, 2015): our result characterizes the "floor" below which no selection mechanism is needed—random bonding alone produces a stable entity ecology without any pressure toward complexity. This establishes the minimal baseline against which objective-free interventions can be measured.

Limitations

Several limitations should be noted. First, although our parameter sweep covers density, grid size, and drift strength, and we additionally test partner-specific rules (§), other rule representations—orientation-dependent rules, rule evolution, and structured sampling—remain untested. Second, the 500-step horizon, while sufficient for stationarity (Figure 9), may miss very slow transients in larger systems. Third, entity observations at each step are not fully independent (the same entity may persist across steps); our significance testing uses all observations but the 0% excess holds identically when restricted to unique types. Fourth, while we supplement the assembly index with automorphism counts and typed-motif census (§), a full multi-metric comparison incorporating compression-based complexity and information-theoretic measures is deferred to future work.

Conclusion

We have presented a systematic application of Assembly Theory to an objective-free artificial life system.

Across 5,000 simulations, 7.1M entity observations, parameter sweeps over 11 conditions, a catalytic positive control, a partner-specific rule expressiveness test, and a motif-gated catalyst control, we find that uniform random bonding rules produce a stable entity ecology but no structurally non-trivial assembly within this rule class. Observed assembly indices are entirely explained by entity size, with zero significant excess over a bond-shuffle null model under both edge-removal and reuse-aware AT formulations, and under standard, partner-specific, and motif-gated catalyst rule tables.

We note explicitly that under the edge-removal formulation, $a_{\text{exact}}(G) = |E(G)|$ is a mathematical identity for connected graphs; the empirical finding is that the reuse-aware formulation—where $a_r(G) < |E(G)|$ when sub-object reuse applies—also shows zero structural specificity. This dual-formulation negative result, confirmed across 7.1M observations, establishes a concrete boundary condition.

This robust negative result—whose reporting is itself valuable given the underrepresentation of negative results in the literature (Fanelli, 2012)—establishes that, within this rule class, emergent structural complexity does not arise from local interactions alone. Structural bias in the rules—whether through adaptation, spatially-oriented conditioning, or structured sampling—appears necessary even without explicit fitness functions. By characterizing this boundary condition precisely, we provide a concrete experimental baseline against which richer rule representations and objective-free interventions can be measured.

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